MIL-P-50578D (AR)
1 May 1985
SUPERSEDING
MIL-P-50578C (AR)
20 June 1979

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549 and M549A1 (WITH WARHEAD AND MOTOR BODY) LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Projectile, 155MM, HE, RA, M549 and M549A1 including loading of the motor body and the warhead.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453 - Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1320

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2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.11)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

9235999 - 155MM, HE, RAP, Assembly, M549
9235998 - Warhead marking and loading assembly HE, 155MM, M549
9327883 - Pallet for Projectile (Marking & Packing)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with the applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scraps from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.5).

3.2 Assembled projectile. The assembled projectile shall comply with all requirements specified on Drawing (dwg) 9235999, including the requirements on dwg. 9235998 for loading and marking the warhead assembly, and with all requirements specified in the applicable specifications and standards.

3.2.1 Motor body. No foreign matter shall be within the interior of the motor body prior to loading of the propellant grains.

3.2.2 Delay. There shall be no lubricant compound on the face of the delay assembly or rocket off cap assembly.

3.2.3 Propellant grains. The surface of the propellant grains shall be free of chips, cracks, gouges, dirt, grease or other foreign material.

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3.2.4 Rotating band. The rotating bands of assembled projectiles shall not have nicks or digs on any land in excess of 1/2 the land width. Surface scratches without evidence of raised copper are acceptable. No nick or digs shall cause the maximum band diameter to be exceeded.

3.3 Cavitation in high explosive charge

3.3.1 Cracks and projected cavities. Cracks and cavities within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.4.3.1. Cavities having a maximum (max) projected length of 1/32 inch (in) or less shall be disregarded.

TABLE I Cast defects.

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks, and annular rings, square inch (sq. in.).	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks, and annular rings (in.).	1/8	1/2	1/2	3/4
Piping cavities, max. projected area (sq. in.).	0	1/4	1/2	--
Piping cavities, max. projected width (in.).	0	1/4	1/4	--
Cracks, max. projected width (in.).	1/32	1/32	1/16	--
Annular rings, max. projected width (in.).	0	0	1/4	--

* If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1 20 square inch. Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

An annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

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3.3.2 Surface cavitation. Cavities in Segment D (see Figure 1) formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well permitted by the drawing, and provided no point extends above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/32 inch or less shall be disregarded in this computation as shall any voids above Segment D).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. Not more than one transverse crack shall be permitted in Segment A and not more than a total of the two transverse cracks shall be permitted in Segment A, B and C combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation. There shall be no base separation in excess of fifteen thousands of an inch (0.015) between the base of the shell cavity and the cast when determined in accordance with 4.5.4.

3.3.6 Metal parts defects. There shall be no cracks, pipes or visually apparent thin wall sections in the warhead (see 4.4.3.1.2).

3.3.7 Foreign material. There shall be no identifiable foreign material in the explosive charge (4.4.3.1.3).

3.4 Loading of high explosive

3.4.1 Mating surfaces. The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the warhead.

3.4.2 Solid explosives. Solid explosives shall not be in the warhead before loading or added to the warhead during initial pouring or cooling.

3.4.3 Coating. Warheads having the interior wall imperfectly coated shall not be loaded. The interior surface of the warhead shall be dry when the high explosive is poured into the warhead (see 4.4.2.1).

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3.4.4 Warhead interior cavity. The interior cavity of the warhead prior to loading shall be free from foreign matter.

3.4.5 Fuze well. The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.4.6 Unsolidified pool of paint. There shall be no unsolidified pool of paint in the bottom of the cavity prior to loading of the explosive.

3.4.7 Exterior surface. There shall be no explosive on the exterior surface of the warhead after loading.

3.5 Assembly torque. The motor body shall be assembled to the warhead while the adhesive is still wet to meet the torque requirement specified on dwg. 9235999. The equipment used for applying torque to the motor body-warhead assembly shall be calibrated at the start of each production shift as well as each time there is a change in equipment, settings or process that is likely to affect uniformity of performance as determined by the Government. The equipment shall be provided with a suitable indicator to aid in observance of proper torque application (see 4.4.3.2).

3.6 Proving ground test. The projectile shall comply with the following requirements when tested as specified in 4.5.3.

3.6.1 Functioning

3.6.1.1 Rocket motor functioning. When the rocket-off cap is removed (rocket motor to function), the rocket shall fire in flight.

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3.6.1.2 Horizontal range.

3.6.1.2.1 Precision. The pooled range probable error (RPE) of the ballistic samples fired as specified in 4.5.3 shall not exceed the appropriate value listed below:

Degrees of freedom	Maximum R (percent)
29	.364
28	.365
27	.366
26	.367
25	.370
9	.441
8	.454
7	.471
6	.495
5	.529

3.6.1.2.2 Minimum range. No projectile shall have a horizontal range less than 97.5 percent of the mean horizontal range.

3.6.1.3 Security. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.6.2 Safety. There shall be no premature burst, rocket motor malfunction or indication of metal parts separation in the gun bore or in flight when fired at any of the conditions specified in 4.5.3. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean horizontal range of its group.

3.7 Inhibitor separation. The propellant grain shall contain no individual separation between the inhibitor and the grain or between the inhibitor layers (delamination) which is greater than 0.01 square inch. The total combined area of propellant separations and inhibitor delaminations shall be no greater than 0.02 square inch. There shall be no propellant inhibitor separation within one inch of any split surface.

3.8 Maximum bourrelet diameter. A ring gage, manufactured to maximum bourrelet diameter shall pass over the warhead and the rocket motor warhead joint after final touch-up painting.

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3.9 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by ARDC for use in radiographing this projectile. A substitute item shall not be used before prior testing and approval by US Army Armament Research and Development Center, Dover, New Jersey 07801-5001.

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, AX-1 or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

GAF Corporation
140 West 51st Street
New York City, New York 10020

Film Types: 800 or 400

3.10 Exudation. After post heat cycling of loaded warheads, there shall be no exudation in the liner, nose threads or exterior of the projectile.

3.11 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.12 Workmanship. The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.12.1 Cleaning. If required, the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.12.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each motor body will not be nicked or burred.

3.12.3 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

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3.12.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Projectile	9235999	62
Warhead Loading Assembly	9235998	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE: Projectile, 155MM, M549		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM, HE, M549/M549A1 Loading, Assy, and Packing (Dwg. 9235999)				
	Examination for defects	62		3.2	4.4.2.12
	Proving Ground Test	60		3.6	4.5.3
	Examination for defects	a.		3.2	4.4.2.1 thru 4.4.2.11
	Warhead Loading Assembly (Dwg. 9235998)				
	Examination for defects	10		3.2	4.4.2.3/4.4.2.4
	Radiographic Inspection	10		3.3	4.4.3.1

NOTES:
a. Two of the projectiles submittted for first article shall be disassembled to allow checking of the examinations from para. 4.4.2.1 through 4.4.2.13.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projectiles shall contain:

a. Warhead metal parts of one lot interfix number from one manufacturer.

b. TNT of the same type from one lot interfix number from one manufacturer; or Composition B from the same grade from the one interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

c. Supplementary charges of one lot interfix number from one manufacturer.

d. Loaded projectile of one weight zone.

e. Motor body metal parts of one lot interfix number from one manufacturer.

f. Motor body forward propellant grains from up to four (4) consecutively produced lots of one lot interfix number from one manufacturer.

g. Motor body aft propellant grains from up to four (4) consecutively produced lots of one lot interfix number from one manufacturer.

h. Delay assemblies of one lot from one manufacturer.

i. Comp B (Grade A or B) explosive used in each ammunition lot, shall consist of Grade B wax from one source only.

4.4.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Warhead Marking and Loading Assembly Prior to Loading	OF 1	9235998
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in charge cavity		100%	3.4.4	Visual
2	Unsolidified pool of paint or blisters in warhead cavity		100%	3.4.6	Visual/Manual
Major					
101	Interior protective coating with bare spots		0.40%	3.4.3	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

11

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Warhead Marking and Loading Assembly (Prior to Assembly of Liner)				9235998
					NEXT HIGHER ASSEMBLY 9235999
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of explosive on exterior of shell body including joint, threads and knurl		100%	3.4.7	Visual
Major					
101	Depth of fuze well		0.40%	3.2	Gage
102	Foreign matter or loose explosive in fuze well		0.40%	3.4.5	Visual
Minor					
201	Diameter of fuze well, max.		0.65%	3.2	Gage
202	Surface cavitation excessive		0.65%	3.3.2	Visual
203	Anti-seize compound not 360 degrees around liner		100%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Warhead Marking and Loading Assembly Prior to Assembly of Supplementary Charge	1 OF 1	9235998
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
Major					
101	True position of inside diameter of liner with minor diameter of internal thread and end face of assembly		0.40%	3.2	Gage
102	Depth to bottom of liner, minimum		0.40%	3.2	Gage
Minor					
201	Liner damaged		0.65%	3.2	Visual
202	Foreign matter in liner cavity including explosive		0.65%	3.2	Visual
203	Liner removable by hand		0.65%	3.2	Manual
204	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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MIL-P-50578D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Warhead Loading Assembly, Prior to Assembly of Lifting Plug	1 OF 1	9235998
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Supplementary charge pad or supplementary charge missing		100%	3.2	Visual
Major					
101	Supplementary charge not removable by hand		0.40%	3.2	Manual
102	Supplementary charge inverted		0.40%	3.2	Visual
Minor					
201	Supplementary charge damaged		0.65%	3.2	Visual
202	Foreign matter on threads including explosive		0.65%	3.4.1	Visual
203	Spacer missing or damaged		0.65%	3.2	Visual
204	Anti-seize compound missing from threads		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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MIL-P-50578D (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.5	Delay Assembly, Prior to Assembly to Motor Body		1	1	9235983
					NEXT HIGHER ASSEMBLY 9235999
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	O-ring missing		0.40%	3.2	Visual
102	Lubricant missing or inadequate on O-ring		0.40%	3.2	Visual
Minor					
201	Evidence or poor workmanship		1.0%	3.12	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.6	Motorbody Prior to Assembly of Delay Assembly		1 1	9235979
				NEXT HIGHER ASSEMBLY
				9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Nozzle insert missing, loose, inverted or damaged		100%	3.2	Visual/Manual
Major	None defined				
Minor	None defined				

NOTES:

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MIL-P-50578D (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.7	Delay Assembly, After Assembly to Motor Body			1 1	9235999
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ring seal missing or not pressed		0.40%	3.2	Visual
102	Lubricant on face of delay assembly		0.40%	3.2.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Rocket-off Cap Assembly, Prior to Assembly to Motor Body	1	1	9276682
				NEXT HIGHER ASSEMBLY: 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rocket-off cap o-ring or lathe cut ring seal missing		100%	3.2	Visual
102	Lubricant not applied to o-ring		100%	3.2	Visual
103	Lubricant on face of rocket-off cap assembly		0.40%	3.2.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.9	Motor Body, Prior to Loading Aft Propellant Grain		1 1		9235999
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Nozzle insert missing, inverted or damaged		100%	3.2	Visual
Major					
101	Foreign matter in cavity		0.40%	3.2.1	Visual
102	Grain spacers missing or obviously improperly assembled		0.40%	3.2	Visual/4.4.3.7
103	Presence of any chips, cracks, gouges, dirt, grease or other foreign material on surface of the aft grain		0.40%	3.2.3	Visual/4.4.3.7
104	Inhibitor separation		0.40%	3.7	Visual/4.4.3.7
Minor					
201	Evidence of poor workmanship		1.0%	3.12 excluding 3.12.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Motor Body, Prior to Loading of Forward Propellant Grain	1	1	9235999
				NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Grain support bonding assembly missing or improperly assembled		100%	3.2	Visual
Major					
101	Aft grain cushion missing prior to assembly of trap		0.40%	3.2	Visual
102	Grain support cushion not properly bonded to trap		0.40%	3.2	Visual
103	Grain spacers missing or obviously improperly assembled		0.40%	3.2	Visual/4.4.3.7
104	Inhibitor separation		0.40%	3.7	Visual/4.4.3.7
105	Presence of any chips, cracks, gouges, dirt, grease or other foreign material on surface of the forward grain		0.40%	3.2.3	Visual/4.4.3.7
Minor					
201	Evidence of poor workmanship		1.0%	3.12 excluding 3.12.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9235999
4.4.2.11	Motor Body and Warhead Prior to Assembly				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Knurl and threads free from grease, explosive, or foreign material, prior to application of sealant		100%	3.2	Visual
Major					
101	O-ring missing from warhead		0.40%	3.2	Visual
102	Warhead insulation missing (or fails to meet pull test load of Dwg. 9235997 if insulator is assembled at load, assembly and packing facility)		0.40%	3.2	Visual/Test
103	Surface primer or sealing compound missing or improperly applied to motor body/warhead threads		0.40%	3.2	Visual
104	Forward propellant grain missing or markings not visible		100%	3.2	Visual
105	Grain cushion missing or lubricant improperly applied		0.40%	3.2	Visual
106	Warhead base free from grease, rust or other foreign material prior to bonding of warhead insulation (applicable only at the load assembly and packing facility)		0.40%	3.2	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9235999
4.4.2.11	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107 108	Lubricant not applied to O-ring Insulation or adhesive above minimum recess (if insulator is assembled at the load, assembly and packing facility)		0.40% 0.40%	3.2 3.2	Visual Gage
Minor 201	Evidence of poor workmanship		1.0%	3.12 excluding 3.12.4	Visual
NOTES:					

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Projectile Assembly, After Final Weighing and Prior to Final Assembly of Lifting Plug	1 OF 1	9235999
			NEXT HIGHER ASSEMBLY -----

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile zone weight incorrect		100%	3.2	Balance
2	Zone stenciling or punch marks missing, inadequate or improper		100%	3.2	Visual
Major	None defined				
Minor 201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9235999
4.4.2.13	Projectile Assembly				NEXT HIGHER ASSEMBLY -----
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Obturating ring missing or damaged		100%	3.2	Visual
Major					
101	Excessive gap between warhead and motor body assembly		100%	3.2	Gage
102	Projectile fails to pass through ring gage		100%	3.8	Gage/4.4.3.5/4.5.5
103	Rotating band nicked or damaged		0.40%	3.2.4	Visual
104	Lifting plug not fully seated		0.40%	3.2	Visual
105	Lifting plug gasket missing		0.40%	3.2	Visual
106	Torque test failure of lifting plug (loosening at min. disassembly torque)		0.40%	3.2	Test/4.5.2
107	Torque test failure of Rocket-off cap (to be checked at the last station prior to palletizing)		0.40%	3.2	Test/4.4.3.8/4.5.2
108	Torque test failure of lifting plug (failure to loosen at max. disassembly torque)		0.40%	3.2	Test/4.5.2
109	Rocket motor and warhead assembly torque		4.4.3.2	3.5	Visual
110	Re-application of assembly torque to rocket motor/warhead joint		4.4.3.2	3.5	Test/4.5.2

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50578D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9235999
4.4.2.13	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY -----
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Painted surface with total exposed or damaged area in excess of 1/4 sq. in.		0.65%	3.2	Visual
202	Marking missing, misleading, or unidentifiable including incorrect color code (see dwg. 9235998 and 9235999)		0.65%	3.2	Visual
203	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
204	Warhead lot number not stamped on motor body rotating band		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Pallet		9327883
			NEXT HIGHER ASSEMBLY -----

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	None defined				
Major					
101	Pallet damaged or improperly assembled		0.40%	3.2	Visual
102	National stock number or DOD Code missing or unidentifiable		0.40%	3.2	Visual
103	Strapping missing or loose		0.40%	3.2	Visual/Manual
104	Projectiles of different weight zones on the same pallet		0.40%	3.2	Visual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping improperly engaged		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and the following:

4.4.3.1 Cavitation. (see 3.3 except 3.3.2). Cavitation defects are classified as follows:

Critical:

1. Cavitation, porosity, piping, cracks or annular rings in Segment A or B in excess of that permitted in Table I.

2. Cavitation, cracks or annular rings in Segment C in excess of that permitted in Table I.

3. Cracks in excess of that permitted by 3.3.4.

4. Base separation (see 4.4.3.1.1).

5. Cavitation in Segment D in excess of that permitted in Table I.

This test shall be performed on all loaded warhead assemblies (100 percent inspection). The radiographic test shall be performed as specified in 4.5.1, using equipment in accordance with 4.4.4. Any assembly which has any one of the above critical defects shall be classed defective and removed from the lot.

Warheads rejected by radiographic examination may be drilled out to one half the depth of the "B" segment in Figure 1 and re-poured, but not more than one drilling operation may be performed per warhead.

4.4.3.1.1 Base separation. Warheads showing the largest base separation as determined by Radiographic examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select warheads at the rate of one (1) out of seven hundred (700) warheads to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These warheads are to be included in the post cycle heat treatment and sectioned for determination of base separation thereafter. If insufficient samples are available with base separation based on radiographs, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning shall be placed in the coolest area of the post cycle igloo. Any warhead after sectioning which has base separation in excess of that permitted by the applicable requirement, shall be classed defective, and the entire post cycle heat treatment lot rejected. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4. Any lot which fails after being post cycle heat treated three times shall be rejected.

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4.4.3.1.2 Metal parts defects (Critical defect). All radiographs of the warhead section loading assemblies shall be examined for evidence of cracks, pipes and visually apparent thin wall sections in the warhead. The presence of any such indication shall cause that assembly to be classed defective and removed from the lot (See section 6.13).

4.4.3.1.3 Foreign material (Critical defect). All radiographs of warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any identifiable foreign material shall cause that assembly to be classed defective and removed from the lot.

4.4.3.2 Motor body - warhead assembly torque. The torque applied to the motor body-warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hours production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

4.4.3.3 Radiographic examination for missing or improperly assembled components - (See dwg. 9235999). This test shall be performed 100 percent. Each assembly shall be subjected to a radiographic examination in a plane that will clearly define all of the Motor Body components, including both channels of the delay assembly. Necessary controls will be implemented to assure that the assembly will be oriented such that both channels are visible when radiographed. The test shall be performed as specified in 4.5.1.2 with equipment in accordance with 4.4.4. Any assembly which shows evidence of missing or improperly assembled components, as listed below, shall be classed as defective and removed from the lot:

a. Delay housing assembly not properly seated in motor body.

b. Shear ring seal not fully expanded in delay and motor body grooves. The groove in the motor body shall be completely filled with lead. There shall be no gap between the shear ring seal and the shoulder of the expansion cone.

c. Missing Boron Potassium Nitrate charge or Delay Columns.

d. Gap between the aft propellant grain and the motor body base exceeding 0.025 inches maximum as compared to the radiographic standard.

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e. Gap between the trap and motor body step exceeding 0.050 inches maximum as compared to the radiographic standard.

f. Gap between the forward propellant grain and the grain support bonding assembly.

g. Evidence of foreign material present.

h. Any other deviation from the normal assembly condition.

4.4.3.4 Proving ground. These tests shall be performed as specified in 4.5.3. Defects shall be classified as follows:

Critical:

1. Premature burst or metal parts separation.
2. Range less than 97.5 percent of mean horizontal range unless the rocket fails to ignite.
3. Rocket motor separation.

Major:

101. Range standard deviation exceeds allowable limit, (See 3.6.1.2.1).
102. Rocket motor fails to ignite.

Minor: None defined.

4.4.3.4.1 Functioning. The functioning test shall be conducted in accordance with 4.5.3.1.

4.4.3.4.1.1 Initial production. For first article or beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria specified herein, and in 4.4.3.4.2.1, 30 sample projectiles shall be selected at random from each lot for this test and fired with the rocket-off cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected.

4.4.3.4.1.2 Subsequent production. After three consecutive lots have complied with 4.4.3.4.1.1, 10 sample projectiles shall be selected at random from each successive lot for this test, and fired with the rocket-off cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet the requirements of this paragraph or 4.4.3.4.2.2 the provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.4.2 Safety. The safety tests shall be conducted in accordance with 4.5.3.2.

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4.4.3.4.2.1 Initial production. For First Article or beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained herein and in 4.4.3.4.1.1, 30 sample projectiles shall be selected at random from each lot for these tests and fired with the rocket-off cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected.

4.4.3.4.2.2 Subsequent production. After three consecutive lots have complied with 4.4.3.4.2.1, 15 sample projectiles shall be selected at random for these tests, and fired with the rocket-off cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet all the requirements of this paragraph or 4.4.3.4.1.2, the provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.5 Ring gage inspection - (Major defect). This test shall be performed as specified in 4.5.5 in accordance with the following plan. At the start of each shift or joint painting operator change this inspection shall be performed on a one hundred (100) percent basis until thirty-two (32) consecutive units have been inspected without failure. After this criteria has been met only those projectiles that require touch-up painting of the bourrelet shall be one hundred (100) percent ring gaged. However, in the event of a failure due to excessive paint on the rocket motor-warhead joint, when ring gaging for touched up bourrelets, those projectiles immediately preceding the rocket motor joint painting failure not inspected shall be inspected until thirty-two (32) consecutive projectiles have been tested without failure. Any projectile that fails for excessive paint on the bourrelet or on the rocket motor joint shall be classed defective and shall be removed from the lot.

4.4.3.6 Exudation - (Minor defect). Eight warheads from the hottest area of the post cycle oven shall be used for the sample. Any warhead with exudation shall be cause for rejection of the post cycle treatment lot.

4.4.3.7 Grain inspection. If the grains are received in packing sleeves which are used for assembling the grains directly into the motor bodies, then visual inspection of the grains shall be conducted without removing the grains from the sleeves (i.e., only exposed surfaces to be inspected).

4.4.3.8 Rocket-off cap. Projectiles received at this inspection station without a rocket-off cap shall not have a cap assembled, but shall be rejected and removed from the lot. Delay assemblies must be extracted and replaced.

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4.4.4 Inspection equipment. Requirements for inspection equipment to perform examinations and tests are specified in the item detail specification and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection equipment designs.

4.5 Methods of inspection

4.5.1 Radiographic examination

4.5.1.1 Determination of warhead defects. The warheads shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the warhead and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of the explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.3 and 4.5.1.4. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as film or processing related or as acceptable metal parts irregularities. In the event an original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location or classification of a defect, the warhead shall be rotated about its longitudinal axis from the original exposure, and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where re-radiographing reveals multiple superimposed cavities each of which individually complies with the requirements. All additional radiographs of a shell for defect conditions will be identified on the film as retakes and stored; traceable to the orginal.

4.5.1.2 Determination of missing or improperly assembled motor body components. In addition to the warhead examination required by 4.5.1.1, the motor body section of the assembled projectile shall be subject to radiographic examination (see 4.4.3.3). In all exposures the radiographic beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination shall be made to determine compliance with the applicable requirements.

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The aiming point of the beam shall be determined based on standard radiographs (see 4.5.1.5) which give most accurate results, and once the particular beam location is established, it shall not be changed unless new radiographic film standards are established and reviewed by the technical agency (see 4.5.1.5). Beam alignment will be established to optimize detection of gap at aft end of propellant grain.

4.5.1.3 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of a warhead so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each warhead shall be used. When radiographing a number of similar warheads the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-00453.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown in Figure 2. For further details on construction of the penetrameter see MIL-STD-00453.

4.5.1.4 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photoelectric step tablet each hour immediately prior to measurements on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARDC. Referenced tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily

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operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.5 Review of radiographic sensitivity and techniques - (See 6.9). The sensitivity of the radiographic techniques shall be reviewed by submitting one radiograph of the warhead standard and a copy of radiographic procedures prior to the start of initial production. In addition, a radiograph of the motor body section of an assembled projectile shall be randomly selected and submitted for review of the sensitivity of the inspection for proper assembly. These radiographs shall be taken and processed using the same techniques as will be used in production. In addition one radiograph of each of the warhead standards and a radiograph of the motor body section shall be submitted for review once a month for the first three months of each contract and every three months thereafter.

4.5.1.6 Operator qualification. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.2 Torque tests. These tests shall be performed on the sample selected in accordance with 4.4.2.13 and 4.4.3.2. a thin line shall be drawn across the joint in question to aid in observing movement. The minimum assembly torque specified on the drawing for the rocket-off cap and the motor body shall be applied in the assembly direction. The minimum and maximum disassembly torques for the lifting plug shall be applied in the disassembly direction.

4.5.3 Proving ground test. This test shall be performed at a Government Proving Ground in accordance with the applicable acceptance test procedure. The samples selected in accordance with 4.4.3.4 shall be fired in a M199 series cannon. The projectiles shall be fuzed with a M557 series PD fuze. Adequate photographic and instrumental coverage shall be provided to assure compliance with all applicable requirements. If projectiles are received with "ballistic sample numbers" from the contractor then the projectiles shall be randomized with regard to test phase and order of firing. The following information pertaining to this test shall be obtained and reported for all phases as specified below.

a. Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers. This information shall be reported on the firing record.

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b. Observable attribute data shall be reported as follows:

1. Premature functioning
2. Metal parts separation
3. Rocket motor malfunction
4. Impact functioning (information only)

c. Variable round-by-round data. Data specified on sheets one and two of DRDAR-QA Form 530 shall be obtained for each ballistic sample. See Figures 3 and 4 for sample forms.

d. Meteorological (Met) data. Met data shall be obtained to an altitude of 14,000 meters at the start of testing (± 15 min.) and at least every two (2) hours throughout the test.

e. Photographic coverage. Photographic coverage shall be adequate to assure compliance with the applicable requirements. Each projectile ballistically tested shall have four lines painted across the warhead/motor body joint (see 6.10). Photographic coverage shall be provided to determine relative movement between warhead and motor body at this joint. Additionally, a second camera shall be positioned to ascertain ignition (burning) of the delay assembly columns.

f. Radar data. Doppler radar (or a suitable alternative with Government approval) shall be obtained for each round fired until radar tracking cannot be maintained. Raw radar tapes shall be maintained a minimum of six months after release of the firing record.

4.5.3.1 Functioning. All test projectiles shall be conditioned and fired at 70 degrees Farenheit (^{o}F) at Charge 8, M203 series propelling charge conditioned at 70°F. In order to minimize the effects of meteorological conditions the test shall be fired in either one (for normal inspection) or three (for tightened inspection) series of 10 rounds each. The first two subgroup for tightened inspection shall be fired at a quadrant elevation of 33 degrees and the last subgroups for tightened inspection or the single group for normal inspection shall be fired at a quadrant elevation of 51 degrees. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that each 10 round series be completed within one (1) hour. Range probable errors shall be calculated for each 10 round subgroup and then a pooled range probable error (for tightened

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inspection) shall be computed for determination of compliance with 3.6.1.2.1.

$$\text{Subgroup RPE(percent)} = RPEj = \sqrt{\frac{X^2 - 1/n\,(X)^2}{n-1}}\ \frac{X\ 100\% \times .6745}{\text{Ave. Range (j)}}$$

$$\text{Pooled RPE(percent)} = RPEp = \sqrt{\frac{RPE_1^2\ (n_1 - 1) + RPE_2^2\ (n_2 - 1) + RPE_3^2 (n_3 - 1)}{DF}}$$

Where X = Individual horizontal ranges within each subgroup

Where DF = Degrees of freedom = n + n ....n(j) - number of subgroups

4.5.3.2 Safety. The projectiles for this test shall be divided into three equal groups. Group 1 shall be conditioned and fired at -65°F at Charge 8, M203 series propelling charge conditioned at -65°F. Group 2 shall be conditioned and fired at 145°F with an M203 propelling charge conditioned at 145°F. Group 3 shall be conditioned at 145°F and fired with a PXR-6297 proof propelling charge conditioned at 145°F. The cold phase (Group 1) shall be fired at a quadrant elevation of 51 degrees and the hot phases (Group 2 and 3) shall be fired at a quadrant elevation of 24 degrees.

4.5.4 Determination of base separation. After heat treatment the samples from each post cycle heat treatment lot shall be cooled to ambient and sectioned in a manner to expose the interface between the cast and the interior base of the warhead at the appropriate center of the base. The warhead shall be positioned nose down or the cast shall be moved as far towards the nose with finger pressure as possible before measuring for base separation between the shell cavity and cast.

4.5.5 Ring gage. After touch-up or joint painting the sample projectiles per paragraph 4.4.3.5 shall be tested by passing through a ring gage manufactured to a maximum diameter. The ring gage is to pass over the bourrelet and rocket motor-warhead joint.

4.5.6 Exudation. The exterior of the warhead shall be visually inspected for exudation. If no exudate is found, remove the lifting plug and supplementary charge and inspect for exudation visually.

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5. PACKAGING

5.1 Packing - Level A. The projectile shall be packed in accordance with dwg. 9327883.

5.2 Marking. Marking shall be in accordance with dwg. 9327883.

6. NOTES

6.1 Intended use. The components and assemblies covered by this specification are intended for use on the M549 and M549A1 RA Projectile.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

6.4 Photographic densities. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York City, New York, has been satisfactory as a source of information on film densities.

6.5 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.6 Inspection lot size. It is noted that the size of inspection lots of components assemblies or item of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantitites provided all of the lot formaton criteria and sampling provisions of this specification are maintained.

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6.7 Proving ground test summary

Test	Sample Size (see 4.4.3.4)		Requirement
Functioning	* 30	10	a. Premature burst or metal parts separation (see 3.6.1.3).
			b. Individual range (see 3.6.1.2.2).
			c. Rocket Motor function (see 3.6.1.1).
			d. Range standard deviation (see 3.6.1.2.1).

*Applicable only for first three lots or tightened inspection.

Test	Sample Size (see 4.4.3.4)		Requirement
Safety (sample divided into 2 groups and fired hot and cold as per 4.5.3.2	* 30	15	a. Premature burst or metal parts separation (see 3.6.2)
			b. Individual range (see 3.6.2).
			c. Rocket motor body separation
TOTAL SAMPLE PROJECTILES REQUIRED	* 60	25	

*Applicable only for first three lots or tightened inspection. Informational-control projectiles are not included.

6.8 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for loaded items provided by the Department of the Army, two copies of all ballitic test data and ammuntiion data cards shall be forwarded to Department of the Army, ARDC, Dover, New Jersey 07801-5001, ATTN: AMSMC-QAR-Q (D) and SMCAR-LCU-SA (D).

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6.9 Radiographic procedures. Radiographic procedures, standards and sample radiographs shall be submitted to the Contracting Officer for review and approval.

6.10 Ballistic test projectiles. All projectiles tested by a government proving ground for ballistic acceptance test shall have four (4) equally spaced lines painted, at the LAP manufacturing facility, across the warhead/motor body joint using Stencil Ink, White No. 37875 per Spec TT-I-1795. These lines shall be approximately one half (1/2) inch wide by six (6) inches long and positioned with three (3) inches on the warhead and three (3) inches on the motor body.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.12 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement QSTAG 605. When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices, if required.

6.13 Metal parts defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.14 Propellant Grains. Rocket motor propellant grains for use in the M549 or M549A1 may either be cast or extruded. For cast propellant grains drawings 9235987 (foward grain) and 9298110 (aft grain) apply. For extruded grains drawings 9324787 (foward grain) and 9324786 (aft grain) apply. The performance of both types of grains is to be considered equal. However, no projectile lot shall contain both extruded and cast propellant grains.

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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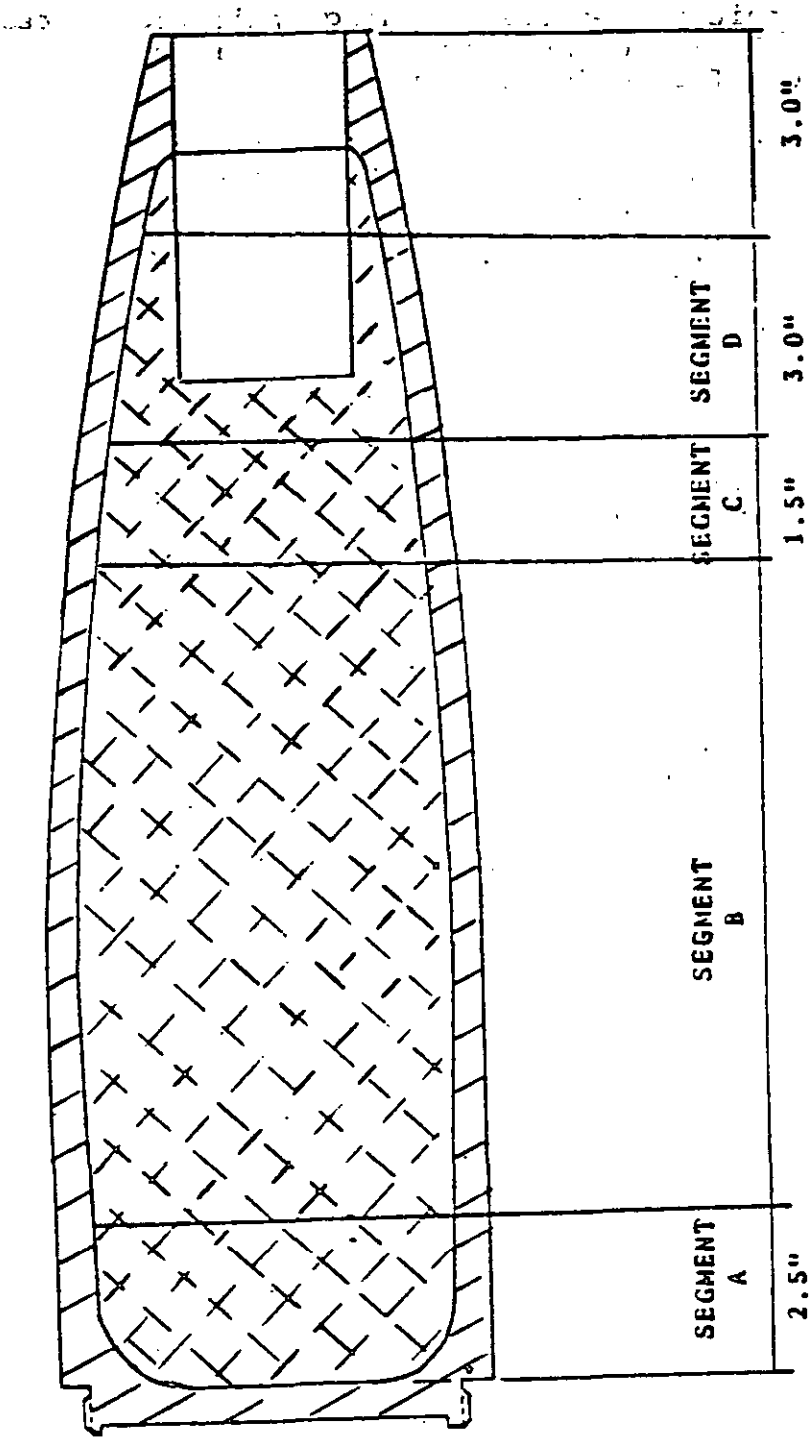
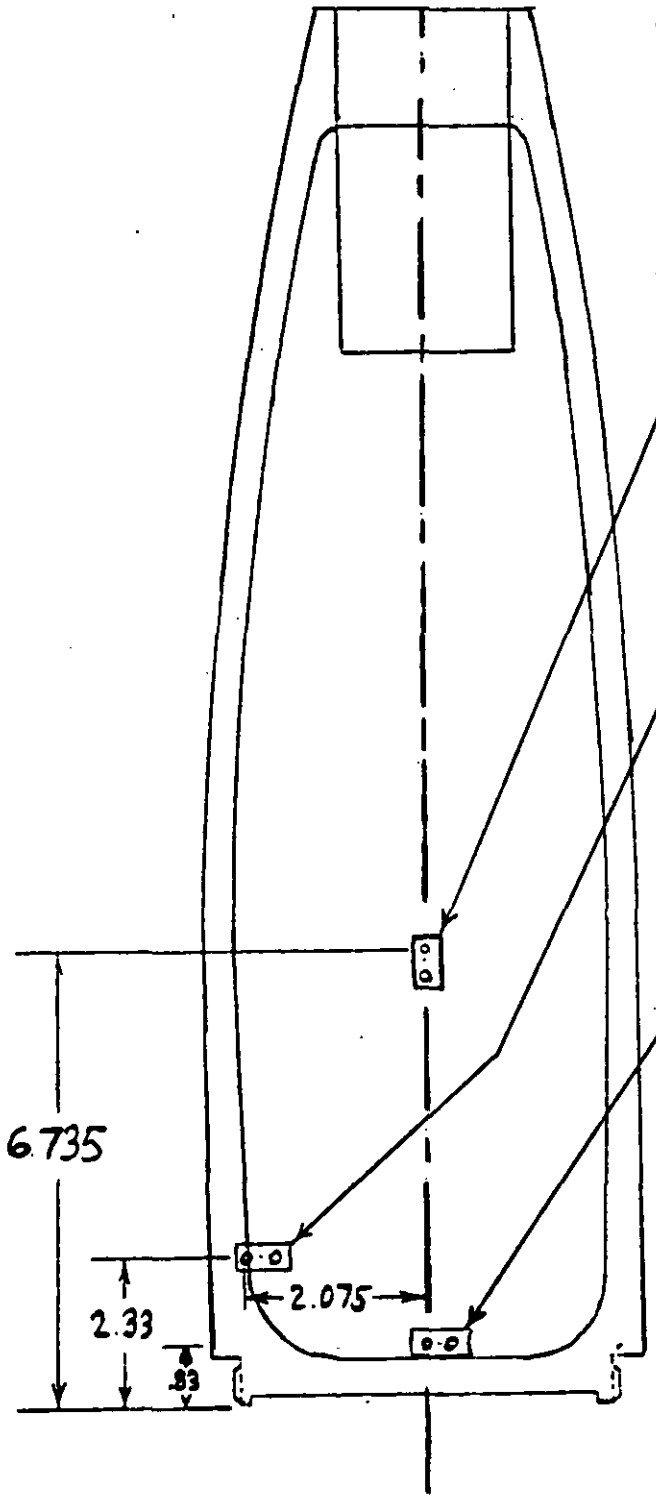


FIGURE 1. Projectile, 155MM, HE, RA, M549 warhead segments.

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Penetrameters constructed per MIL-STD-453.
Dimensions shown are for location of center of 2T holes.

0.385" single steel wall plus 5.32" explosive
Penetrameter thickness = .035"
2T hole diameter = .070"

Position 2T hole on central axis of shell

1.07" single steel wall plus 1.55" explosive at .120" from tangent point
Penetrameter thickness = .050"
2T hole diameter = .100"
Position 2T hole so as tangent to the cavity wall.

1.45" single steel wall plus 3.19" explosive
Penetrameter thickness = .070"
2T hole diameter = .140
Position 2T hole so as tangent to the cavity base and on the central axis of shell.

PROJECTILE, 155MM, HE, RA M549
RADIOGRAPHIC PENETRAMETER REQUIREMENTS

FIGURE 2

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Sheet 1

M549 FIRING RECORD INFORMATION

OVERALL DATA

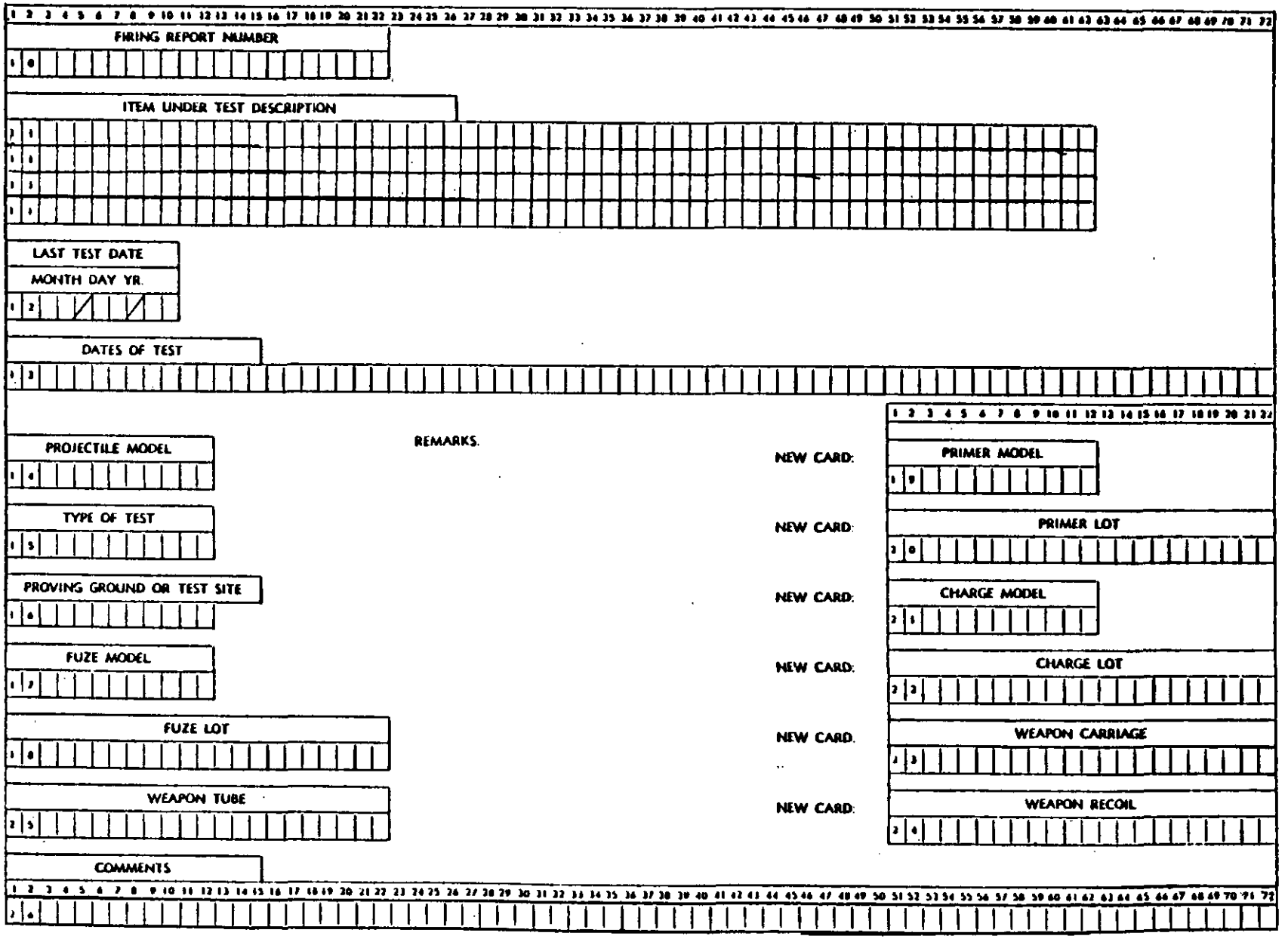


FIGURE 3. Firing record, overall data.

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Sheet 2

M549 FIRING RECORD INFORMATION FOR FIRING RECORD NUMBER

FILL IN

NOTE: A NEW SHEET 2 MUST BE STARTED EACH TIME THERE IS A CHANGE IN QUADRANT ELEVATION, PROJECTILE TEMPERATURE, PROPELLANT TEMPERATURE, DATE FIRED OR PROJECTILE LOT.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Enter different letter code for each Firing Data Form (A, B, etc.) for same firing record

0 1

PROJECTILE LOT

3 9

LOT (LEAVE BLANK)

0 2

QUADRANT ELEVATION (DEGREES)

4 1 .

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

DATE OF FIRING: MONTH - DAY - YEAR

4 0 / /

PROJECTILE TEMPERATURE DEGREES F.

4 3 -

PROPELLANT TEMPERATURE DEGREES F.

4 4 -

TYPE CARDS ONLY WHERE THERE ARE ENTRIES

	TUBE RD. NO.	SAM. NO.	PROJECTILE WEIGHT	TIME FIRED	MUZZLE VEL. (MPS)	AVG. CH. PR. (PSI X 100)	RKT. IGN. (SECS.)	HORIZONTAL RANGE (M)	DEFLECTION (M)
1 2	3 4 5 6	7 8	9 10 11 12 13	14 15 16 17	18 19 20 21 22 23	24 25 26 27 28	29 30 31 32 33	34 35 36 37 38	39 40 41 42 43
0 0			.		.		.		
0 0			.		.		.		
0 0			.		.		.		
0 0			.		.		.		
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COMMENTS AND REMARKS:

DRDAR-QA FORM 530 JUN 78

MIL-P-50578D

FIGURE 4. Firing record.

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Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A719)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-50578D	PROJECTILE, 155MM, HE, RA, M549 AND M549A1 LAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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